



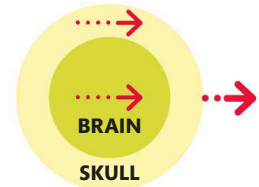
Head injuries

Minor bumps to the head or injuries to the scalp alone have no long-term consequences. However, injury to the brain is potentially extremely serious and can be fatal.

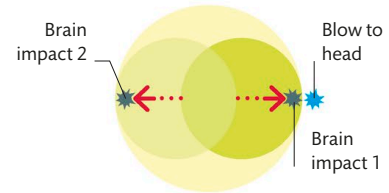
Direct damage to the brain may occur if both the scalp and the skull are penetrated. Indirect damage occurs as a result of a blow to the head that does not damage the skull. In both cases, head injuries can rupture blood vessels, causing a brain hemorrhage. Minor head injuries typically produce only mild, short-lived symptoms, such as a bruise. In some cases, concussion may follow, and this may cause confusion, dizziness, and blurred vision, which may last for several

days. Postconcussive amnesia can also occur. Repeated concussions lead to detectable brain damage, such as impaired cognitive abilities, tremors, and epilepsy.

Severe head injury may produce unconsciousness or coma and usually brain damage. In nonfatal cases, the effects of brain damage may include weakness, paralysis, poor memory and/or concentration, intellectual impairment, and even personality changes. Such effects can be long-term or permanent.



- 1 Rapid movement**
When a person is moving rapidly—for example, on a bike or in a car—the skull and brain are moving at the same speed.



- 2 Stopping suddenly**
On impact, the brain smashes into the front of the skull, rebounds, and sustains further injury as it hits the back of the skull.

Epilepsy

Ranging from mild to life-threatening, epilepsy is a brain function disorder in which there are recurrent seizures or periods of altered consciousness, caused by abnormal electrical activity in the brain.

Often the cause of epilepsy is unknown, but in some cases, it may be due to a brain condition such as tumor or abscess, a head injury, stroke, or a chemical imbalance. Seizures (fits) may be generalized or

partial, depending on how much of the brain is affected by abnormal electrical activity. There are several types of seizures. In a tonic-clonic (grand mal) seizure, the body stiffens before uncontrolled

movements of the limbs and body begin, lasting up to several minutes. In absence (petit mal) seizures, the victim loses consciousness, although muscle is retained.

THERE ARE ABOUT
60 TYPES OF
EPILEPTIC SEIZURES

